Enhancement One Narrative: Software Design/Engineering

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Artifact Description

The artifact I chose is a text-based adventure game I created early in the Computer Science program. It is a Python project where the player moves between rooms, collects gemstones, and tries to escape a villain. I created the original version at the beginning of my degree, when I was still learning basic programming skills. The old version had simple movement and item logic, but the code was not very organized, and everything happened in one long section.

Justification for Inclusion

I included this artifact in my ePortfolio because the updated version clearly shows improvement in both software design and engineering. Even though the game itself is simple, the way I rewrote it shows that I can plan, organize, and structure code in a more professional way. Some enhanced parts that show my skills include:

- I moved the whole game into a Game class so that all the game's data and functions are in one place.
- I broke the code into smaller methods, which makes the program easier to read and change.
- I added command handling through a parse_and_execute method, so the game responds to input in a clean and organized way.
- I removed the hardcoded number of stones and now calculate the total automatically from the room map.
- I added simple error handling, a help command, and a smoother quit option.
- I cleaned up the comments and used clearer naming to help readability.

These improvements show my ability to use better software engineering techniques and make the code easier to maintain and expand.

Course Outcome Connection

In Module One, I planned to have this enhancement support the course outcome related to software design and engineering. I believe I met this goal because I used real software design skills when improving the code. I organized the project in a way that follows better programming practices, reduced repeated code, added clarity, and made the structure easier to understand. I do not think that I needed to update my outcome coverage plan, because this enhancement still fits the outcome I originally planned for.

Reflection on the Enhancement Process

Working on this enhancement showed me how much I have learned since making the original version. When I looked back at the older code, I could see that it worked, and that for what it was intended it was good, but it was not well structured. Turning it into a class-based design required me to think through how the game actually works and how the pieces should fit together. Breaking the program into smaller methods also helped me think like a software engineer, not just someone writing code to make something run.

One challenge I faced was keeping the game the same while still improving the design. I had to make sure that movement, item collection, and the villain room all worked the same way while being rewritten in a cleaner structure. Another challenge was making the command system easier to use. The old version had several long if statements, but the new version uses a command parser that is simple and easier to update.

Overall, this enhancement helped me practice skills such as code organization, readability, input handling, and planning. It also helped me see the progress I have made in the program, which I am proud of. This is why I feel this artifact belongs in my ePortfolio, it shows real improvement in how I think about software design and how I write code today.